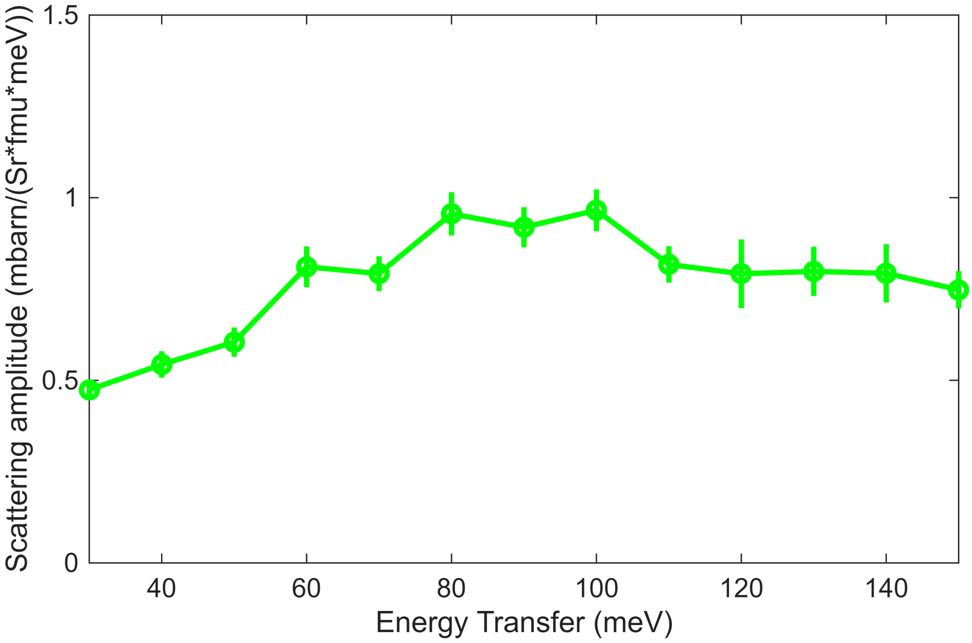
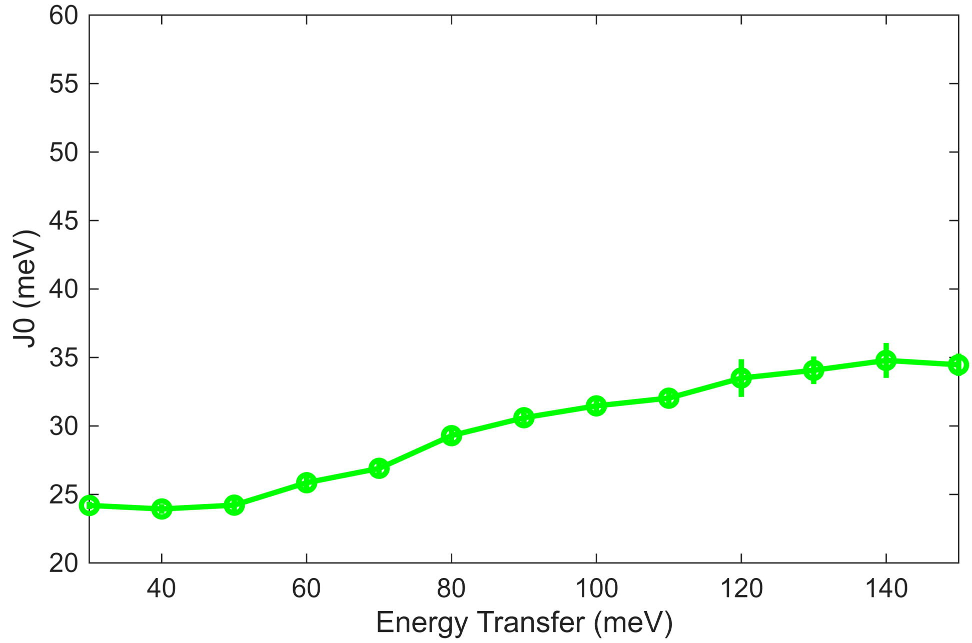
```
fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
%fit_data_file = 'FitEn_cut200_1Dsbg_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
fit_data_file =
'FitEn_cut200ref200_2D_constFFCut_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.
%fit_data_file = 'FitEn_cut200ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly
2-exponent fit fixed bg at high momentum.

if ~exist('fit_resEi200','var')
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi200')
        fit_resEi200 = ld.fit_resEi200;
    else
        fit_resEi200 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei200meV.mat';
if ~exist('cuts2fit200','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit200 = ld.cuts2fit200;
end
cuts2fit = cuts2fit200;
fit_res = fit_resEi200;
mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
%cl = {cuts2fit200.cut110_sel};
%cl = {cuts2fit200.cut200_sel};
cl = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[])};
```

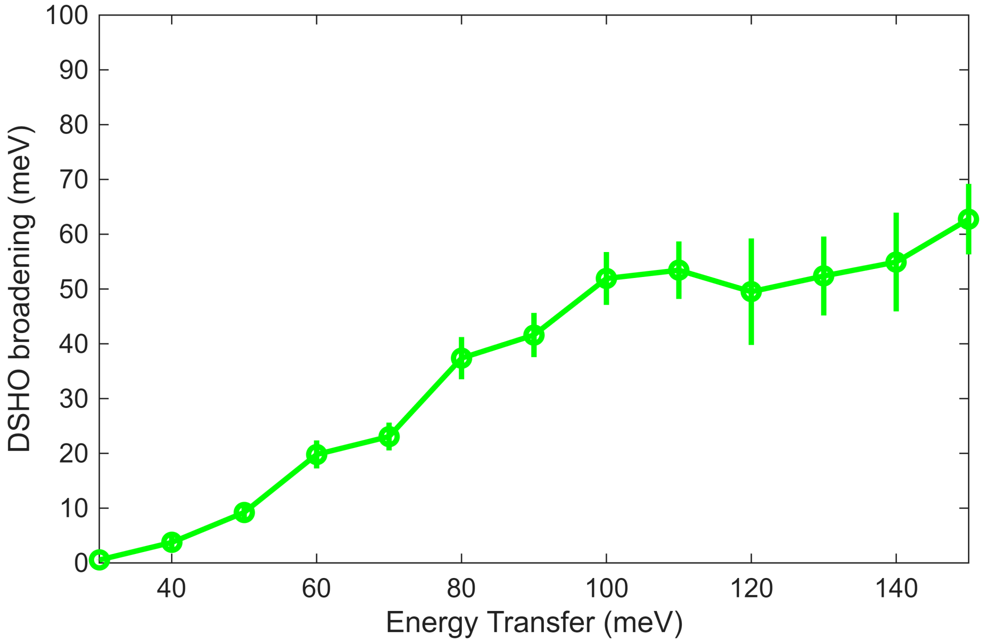
```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  4690699 pixels -- processing data... -----> retained   4604422 pixels
--------------------------------------------------------------------------------
Number of points in input object: 8259853
     Fraction of object selected:    56.79% (=   4690699 points) ;  Access speed:     1.87GB/sec
     Fraction of object retained:    55.74% (=   4604422 points)

**** Total time in cut(sqw)   :      1.0sec Including:
     Data access time fraction:  17.7%; Transf. time frac.:  50.6%; Resulting pix preparation:  31.6%
--------------------------------------------------------------------------------
```

```
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end

the2Dcuts = cl;
```

```
[Sgnl,J0,G] = plot_SJG_results(fit_res,'g',{'S','J0','gamma'},{[0,1.5],[20,60],
[0,100]});
```


Scattering amplitude (mbarn/(Sr*fmu*meV))
1.5
1
0.5
0
40
60
80
100
120
140
Energy Transfer (meV)


J0 (meV)
60
55
50
45
40
35
30
25
20
40
60
80
100
120
140
Energy Transfer (meV)




```
dE = min(Sgnl.x(2:end)-Sgnl.x(1:end-1));
e_min = Sgnl.x(1);
```

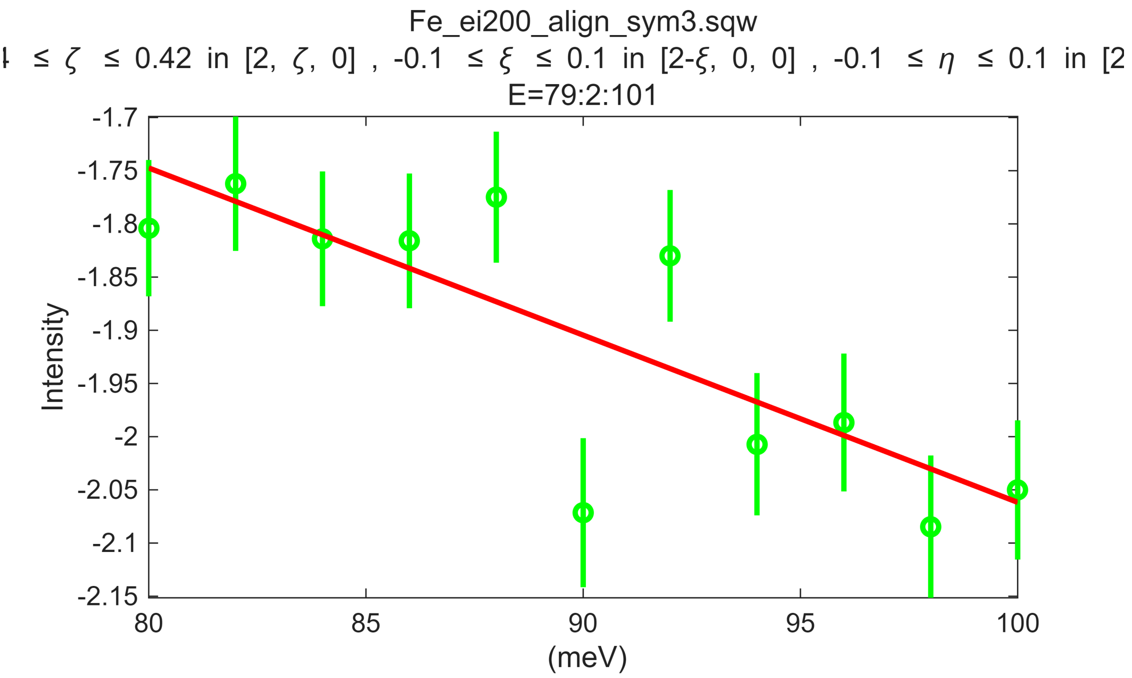
```
En = 90
```

```
En =
90
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10;
en_idx =  floor((En-e_min)/dE)+1;
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);
%the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[])};%(1:2);

%init_bg_params =
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg
%in case of 2-D fitting
%
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p
arams,init_bg_params,false)
lfit_par = fit_res.all_fit_par(en_idx);
en_range = lfit_par.en_range;
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2
,true,false);
```

```
Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.71301
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for    449588 pixels -- processing data... -----> included    405176 pixels
--------------------------------------------------------------------------------
Number of points in input object: 449588
     Fraction of object selected:   100.00% (=    449588 points) ;  Access speed:     1.23GB/sec
     Fraction of object retained:    90.12% (=    405176 points)
```

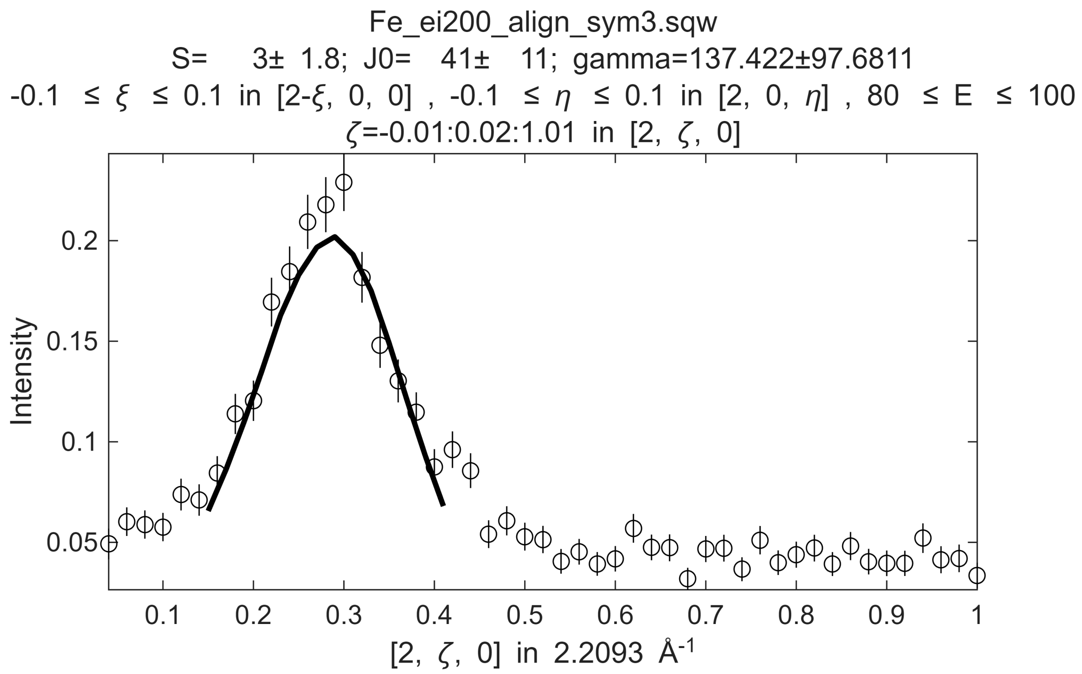
```
**** Total time in cut(sqw)   :      0.1sec Including:
     Data access time fraction:  43.3%; Transf. time frac.:  52.4%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   148916 pixels -- processing data... -----> retained    139254 pixels
--------------------------------------------------------------------------------
Number of points in input object: 452915
     Fraction of object selected:    32.88% (=    148916 points) ;  Access speed:     1.06GB/sec
     Fraction of object retained:    30.75% (=    139254 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  20.3%; Transf. time frac.:  34.8%; Resulting pix preparation:  39.5%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   139254 pixels -- processing data... -----> included    139254 pixels
--------------------------------------------------------------------------------
Number of points in input object: 139254
     Fraction of object selected:   100.00% (=    139254 points) ;  Access speed:     1.75GB/sec
     Fraction of object retained:   100.00% (=    139254 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  34.2%; Transf. time frac.:  22.9%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
```



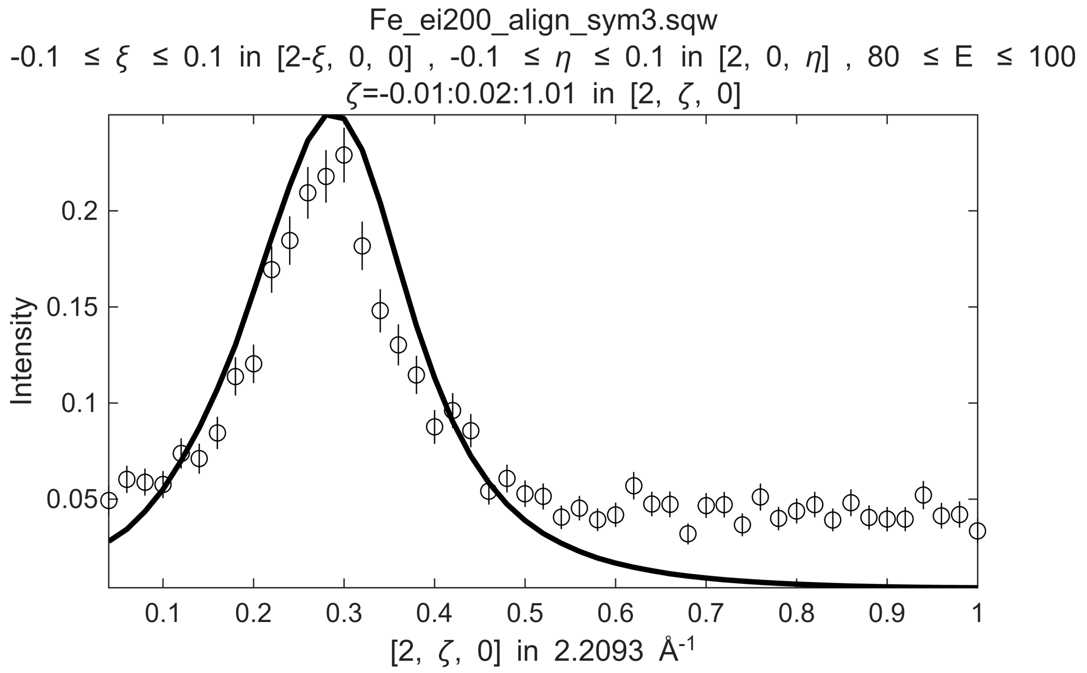

```
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.6888s    Reduced Chi^2 = 15.2893
 Free parameter values:
          41.6         0.9192          30.6              0
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 7.999s    Reduced Chi^2 = 1.2787      Levenberg-Marquardt = 0.1
 Free parameter values:
         37.54         0.7456         29.91       -0.09989
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 14.3205s    Reduced Chi^2 = 1.2205      Levenberg-Marquardt = 0.01
 Free parameter values:
         50.69          1.086            31        -0.1291
--------------------------------------------------------------------------------
Iteration = 3
```

```
------------------
 Total time = 20.6268s    Reduced Chi^2 = 1.2126      Levenberg-Marquardt = 0.001
 Free parameter values:
         74.89          1.669          33.81        -0.1574
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 26.8917s    Reduced Chi^2 = 1.1958      Levenberg-Marquardt = 0.0001
 Free parameter values:
         108.6           2.46          38.06          -0.18
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 33.1485s    Reduced Chi^2 = 1.1737      Levenberg-Marquardt = 1e-05
 Free parameter values:
           135          2.993          41.21        -0.1865
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 39.5607s    Reduced Chi^2 = 1.1744      Levenberg-Marquardt = 1e-06
 Free parameter values:
         143.5           3.13           42.1        -0.1872
 Total time = 41.2058s    Reduced Chi^2 = 1.1738      Levenberg-Marquardt = 1e-05
 Free parameter values:
         143.4          3.129          42.09        -0.1871
 Total time = 42.817s    Reduced Chi^2 = 1.1765      Levenberg-Marquardt = 0.0001
 Free parameter values:
         142.7          3.115          42.01        -0.1869
 Total time = 44.3781s    Reduced Chi^2 = 1.1732      Levenberg-Marquardt = 0.01
 Free parameter values:
         137.4          3.016          41.39        -0.1854
--------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
--------------------------------------------------------

Foreground parameter values:
----------------------------
    1             1
    2             8
    3         137.4   +/-   97.68
    4         3.016   +/-   1.837
    5             0
    6         41.39   +/-   11.3
    7             0
    8             0
    9             0
   10             0


Background parameter values:
----------------------------
    1        0.6125
    2      -0.01571
    3       -0.1854   +/-   0.03028


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9954    0.9987   -0.9496
    0.9954    1.0000    0.9941   -0.9739
    0.9987    0.9941    1.0000   -0.9474
   -0.9496   -0.9739   -0.9474    1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   452915 pixels -- processing data... ----->  included    411177 pixels
--------------------------------------------------------------------------------
```

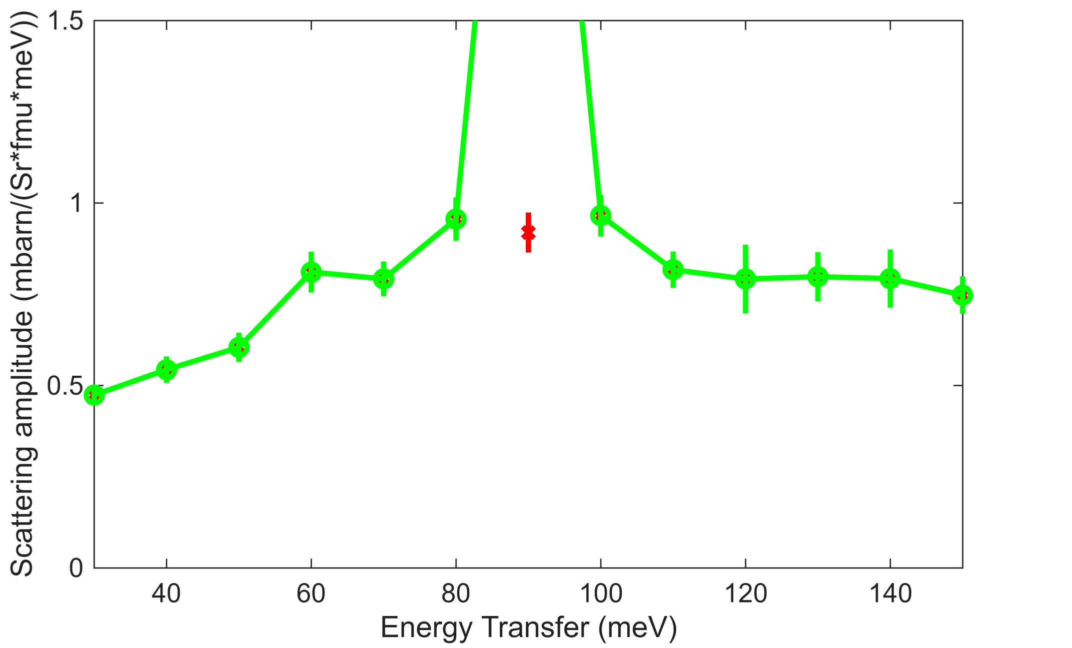
```
Number of points in input object: 452915
     Fraction of object selected:   100.00% (=    452915 points) ;  Access speed:     1.96GB/sec
     Fraction of object retained:    90.78% (=    411177 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  54.7%; Transf. time frac.:  43.3%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   139254 pixels -- processing data... ----->  included    126284 pixels
--------------------------------------------------------------------------------
Number of points in input object: 139254
     Fraction of object selected:   100.00% (=    139254 points) ;  Access speed:     1.45GB/sec
     Fraction of object retained:    90.69% (=    126284 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  34.9%; Transf. time frac.:  62.2%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
```

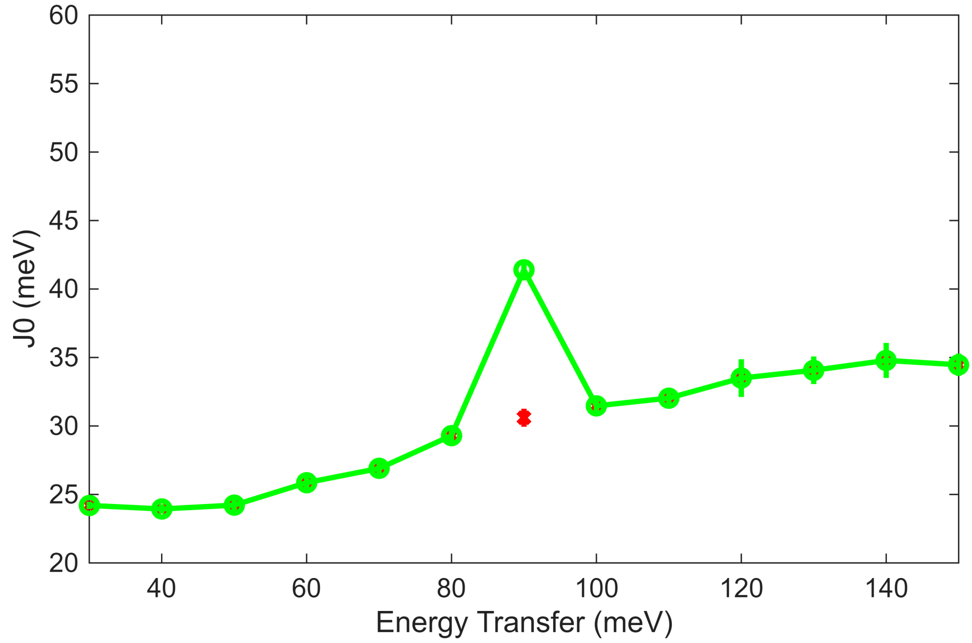


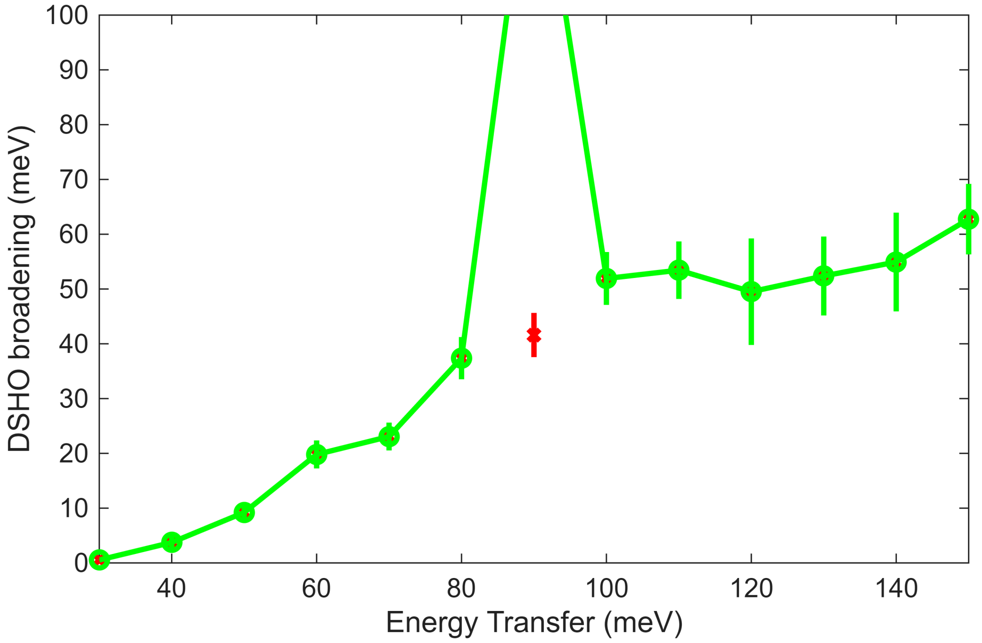

```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   452915 pixels -- processing data... ----->  retained    411177 pixels
--------------------------------------------------------------------------------
Number of points in input object: 452915
     Fraction of object selected:   100.00% (=    452915 points) ;  Access speed:     1.88GB/sec
     Fraction of object retained:    90.78% (=    411177 points)

**** Total time in cut(sqw)   :      0.1sec Including:
     Data access time fraction:  22.6%; Transf. time frac.:  51.1%; Resulting pix preparation:  25.5%
--------------------------------------------------------------------------------
```




```
%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
Sgnl.signal(en_idx) = fit_par.p(4);
J0.signal(en_idx) = fit_par.p(6);
G.signal(en_idx) = fit_par.p(3);
plot_SJG_results(fit_res,'g',{'S','J0','gamma'},{[0,1.5],[20,60],[0,100]},
{Sgnl,J0,G});
```



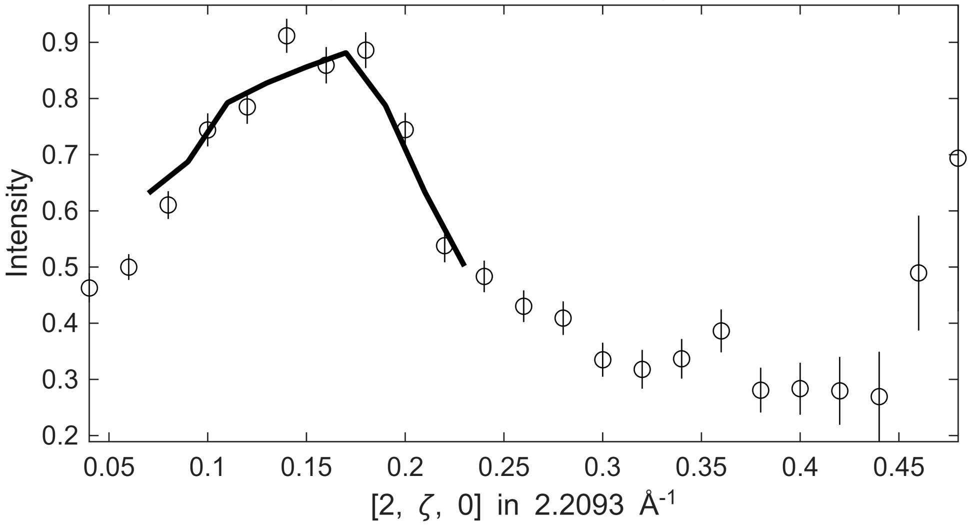




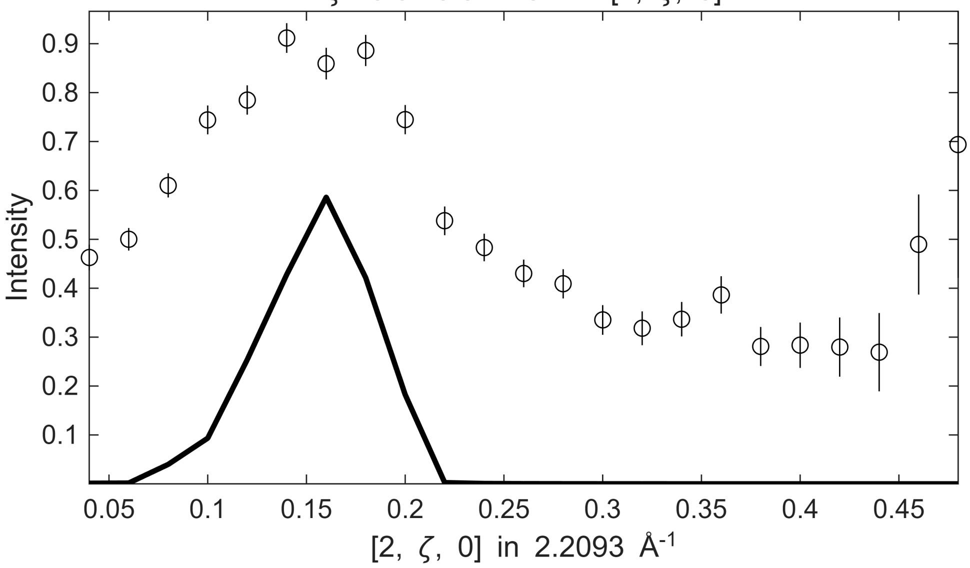

```
[fit_res,Sg,J0,G] = refit_data_around_peaks(the2Dcuts,fit_res,0.2,true);
```

```
*************************************************************
*** step: 1#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*************************************************************
```

Fe_ei200_align_sym3.sqw
S=0.38±0.03; J0=   25±0.072; gamma=0.120188±0.0181987
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 20 ≤ E ≤ 40
ζ=-0.01:0.02:1.01 in [2, ζ, 0]

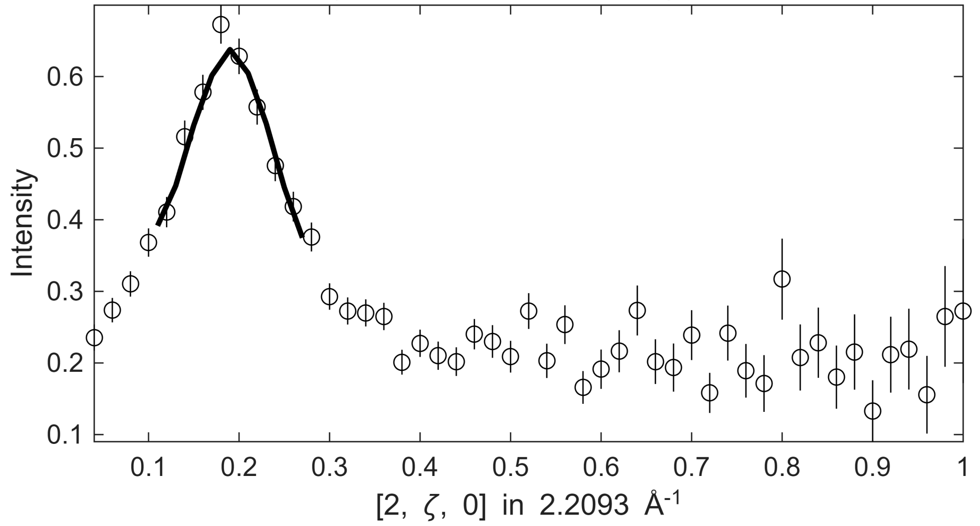


Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 20 ≤ E ≤ 40
ζ=-0.01:0.02:1.01 in [2, ζ, 0]

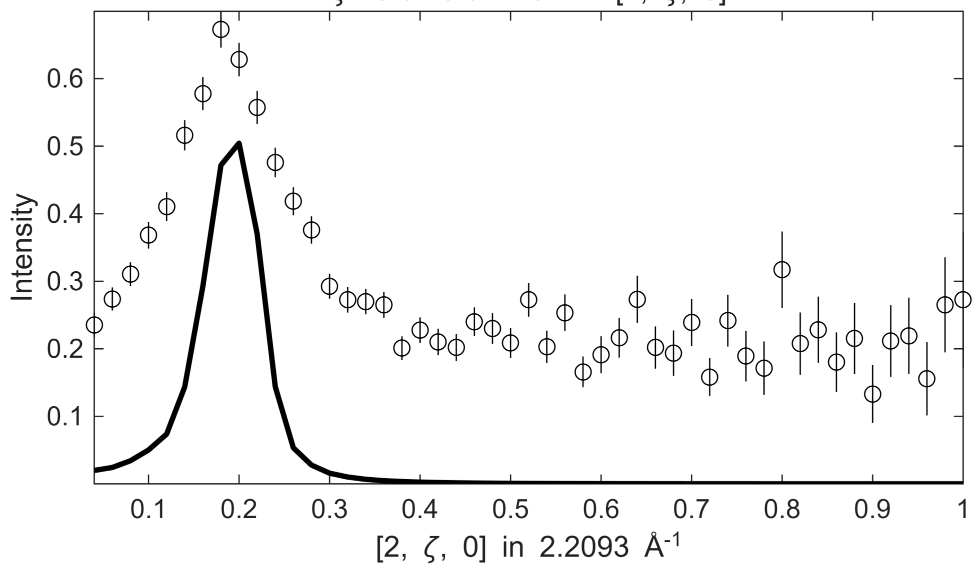



```
**************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
**************************************************************
**************************************************************
*** step: 2#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
**************************************************************
```

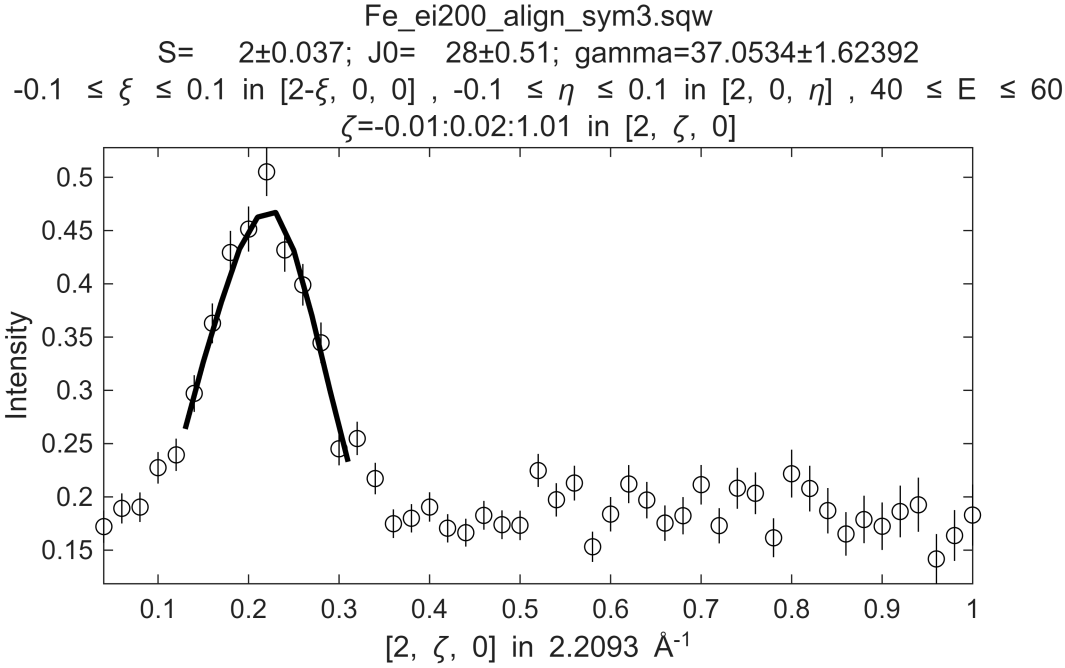
Fe_ei200_align_sym3.sqw
S=0.58±0.019; J0=   24±0.38; gamma=5.88925±0.411297
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 30 ≤ E ≤ 50
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



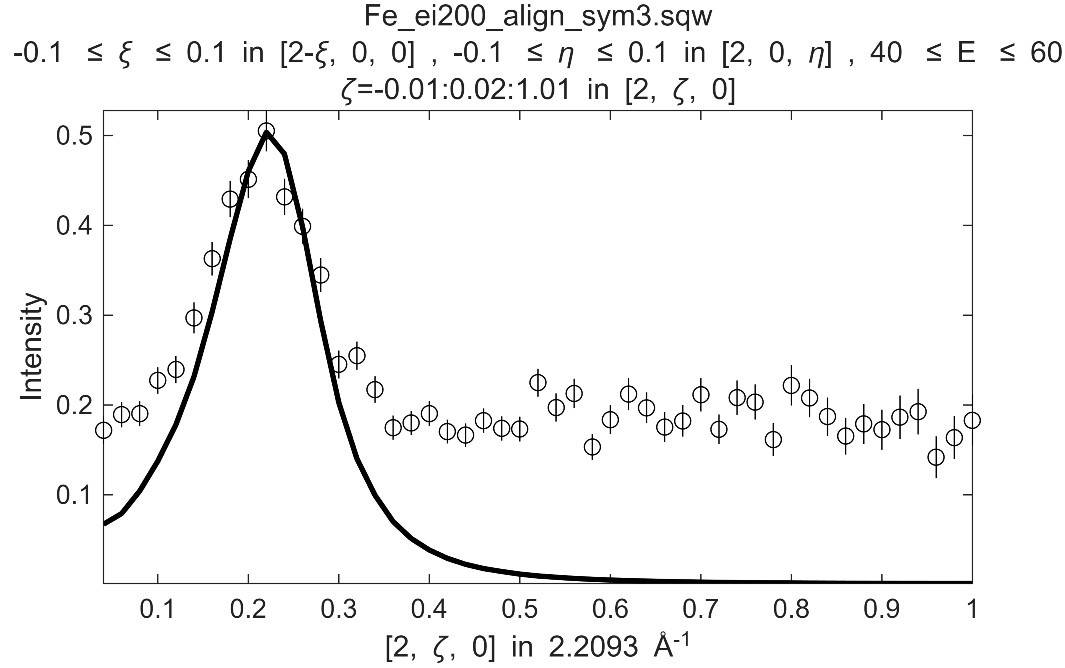
Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 30 ≤ E ≤ 50
ζ=-0.01:0.02:1.01 in [2, ζ, 0]




```
***************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
***************************************************************
***************************************************************
*** step: 3#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
***************************************************************
```

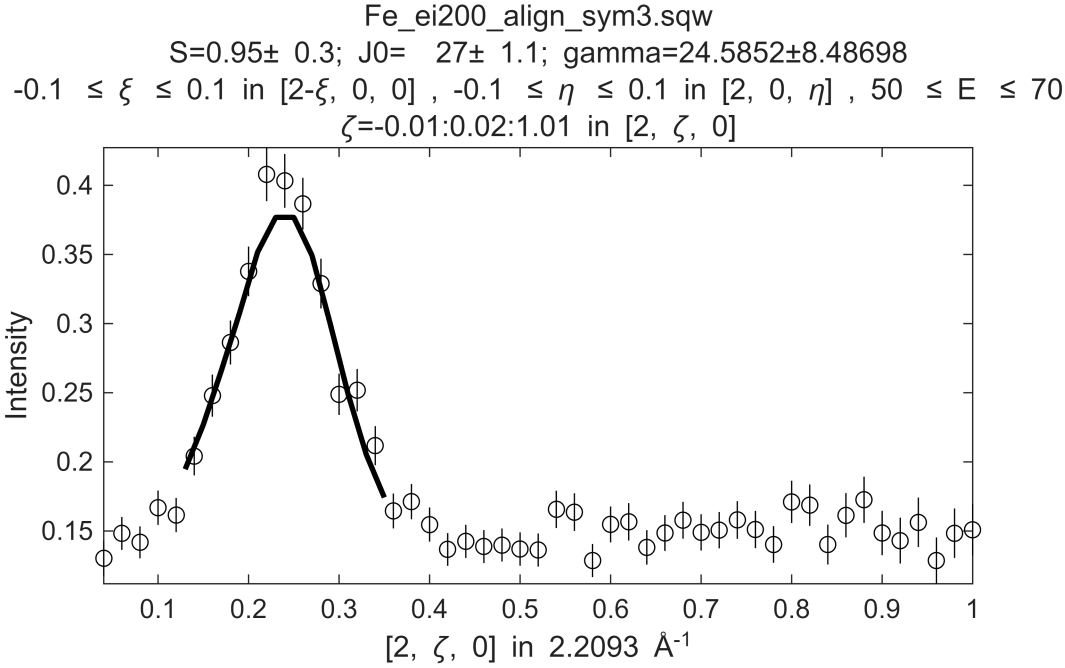


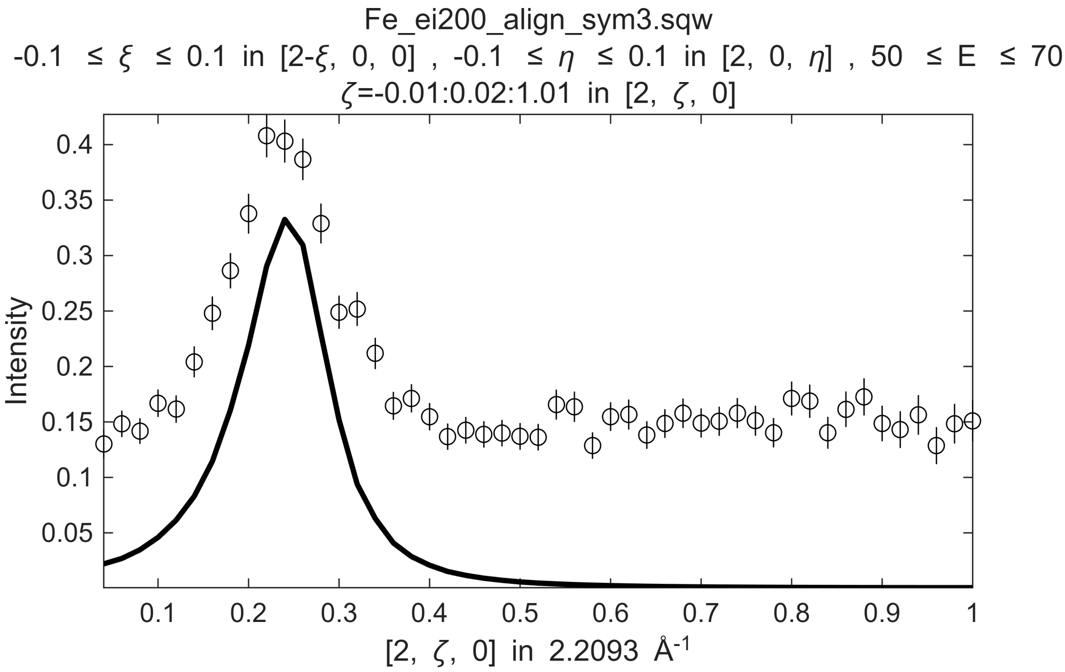
Fe_ei200_align_sym3.sqw
S= 2±0.037; J0= 28±0.51; gamma=37.0534±1.62392
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 40 ≤ E ≤ 60
ζ=-0.01:0.02:1.01 in [2, ζ, 0]
Intensity
[2, ζ, 0] in 2.2093 Å-1



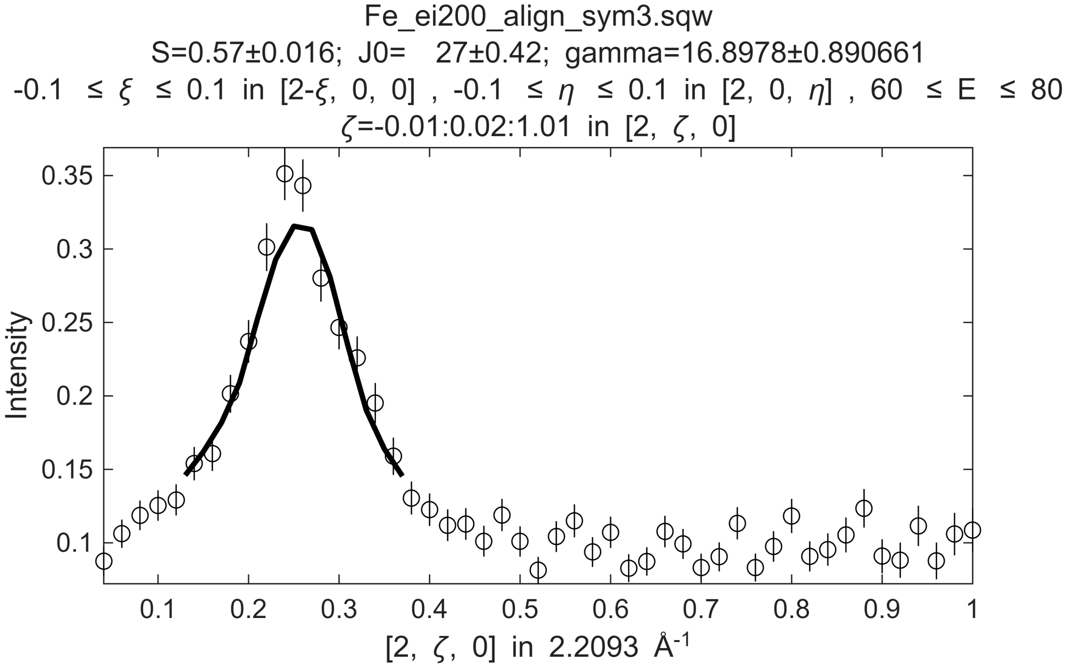
Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 40 ≤ E ≤ 60
ζ=-0.01:0.02:1.01 in [2, ζ, 0]
Intensity
[2, ζ, 0] in 2.2093 Å-1

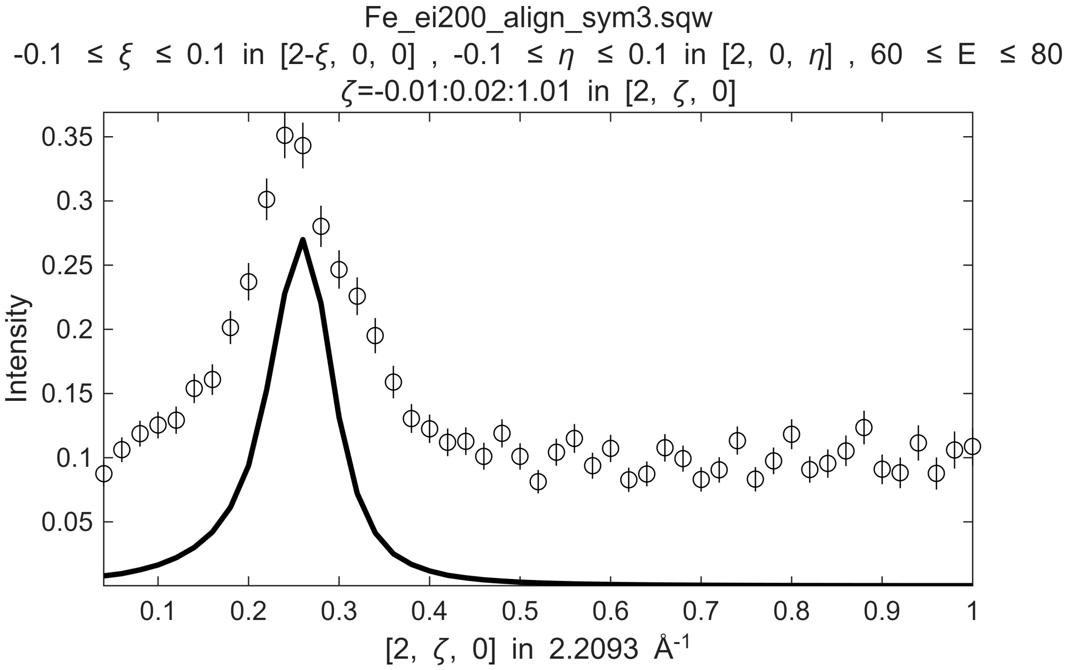

```
*************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*************************************************************
*************************************************************
*** step: 4#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*************************************************************
```


Fe_ei200_align_sym3.sqw
S=0.95± 0.3; J0= 27± 1.1; gamma=24.5852±8.48698
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 50 ≤ E ≤ 70
ζ=-0.01:0.02:1.01 in [2, ζ, 0]
Intensity
[2, ζ, 0] in 2.2093 Å^-1

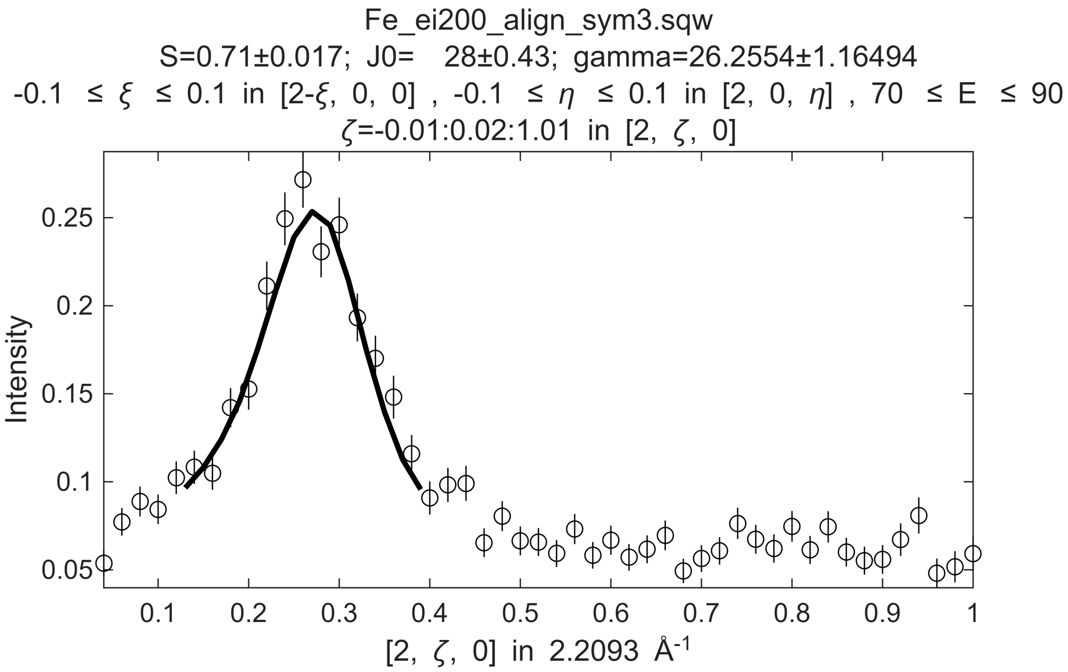

Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 50 ≤ E ≤ 70
ζ=-0.01:0.02:1.01 in [2, ζ, 0]
Intensity
[2, ζ, 0] in 2.2093 Å^-1

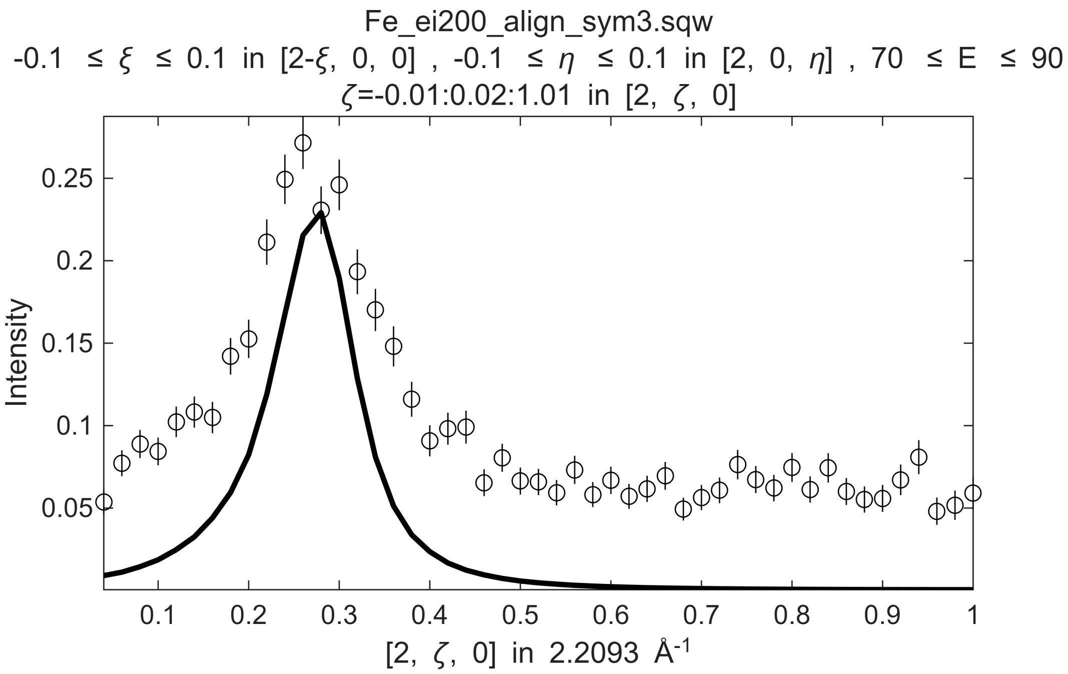
```
***********************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
***********************************************************
***********************************************************
*** step: 5#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
***********************************************************
```


Fe_ei200_align_sym3.sqw
S=0.57±0.016; J0= 27±0.42; gamma=16.8978±0.890661
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 60 ≤ E ≤ 80
ζ=-0.01:0.02:1.01 in [2, ζ, 0]
Intensity
0.35
0.3
0.25
0.2
0.15
0.1
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
[2, ζ, 0] in 2.2093 Å-1


Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 60 ≤ E ≤ 80
ζ=-0.01:0.02:1.01 in [2, ζ, 0]
Intensity
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
[2, ζ, 0] in 2.2093 Å-1

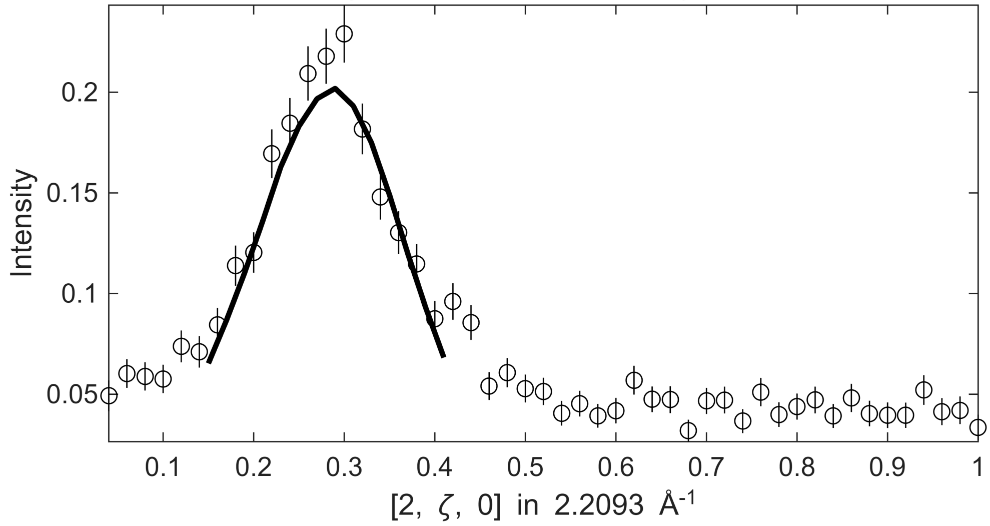
```
************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
************************************************************
************************************************************
*** step: 6#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
************************************************************
```



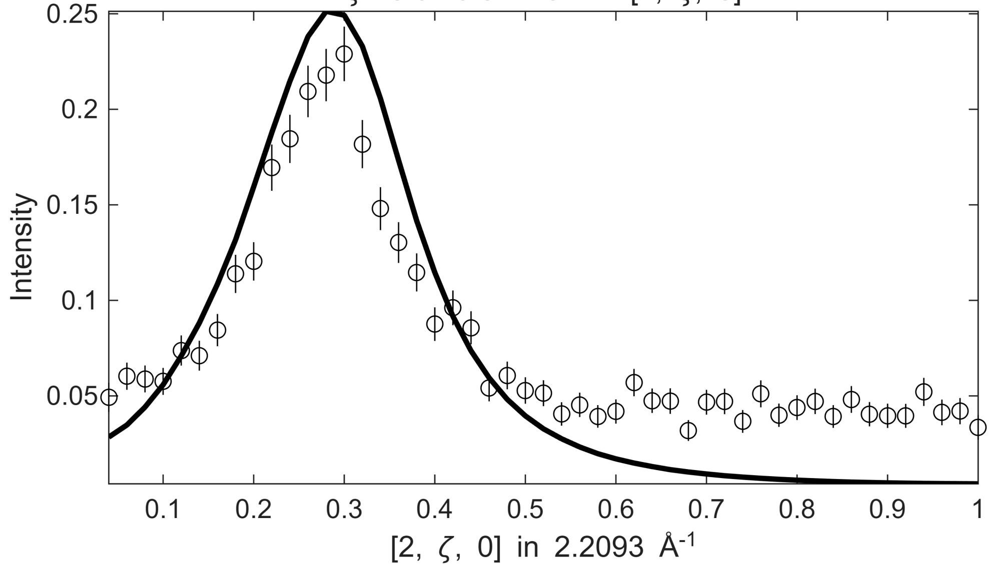



```
**************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
**************************************************************
**************************************************************
*** step: 7#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
**************************************************************
Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.71301
```

Fe_ei200_align_sym3.sqw
S= 3.1±0.068; J0= 42±0.88; gamma=141.691±6.28407
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 80 ≤ E ≤ 100
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 80 ≤ E ≤ 100
ζ=-0.01:0.02:1.01 in [2, ζ, 0]




```
*************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
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*** step: 8#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:8 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.426
```

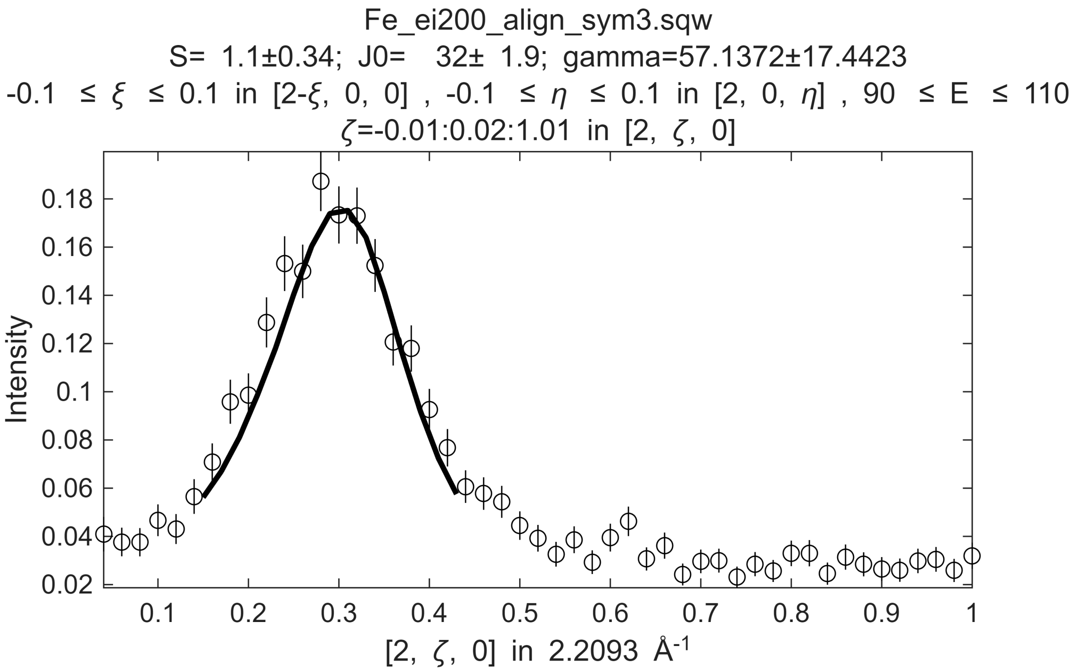

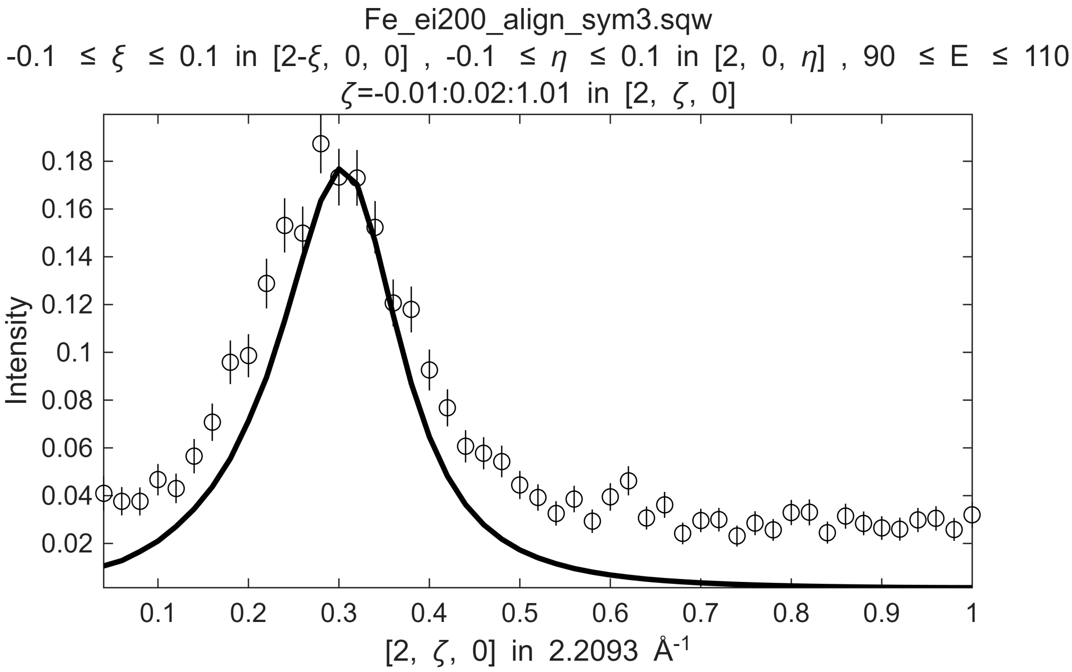


```
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*** step: 9#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:12 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.139
```

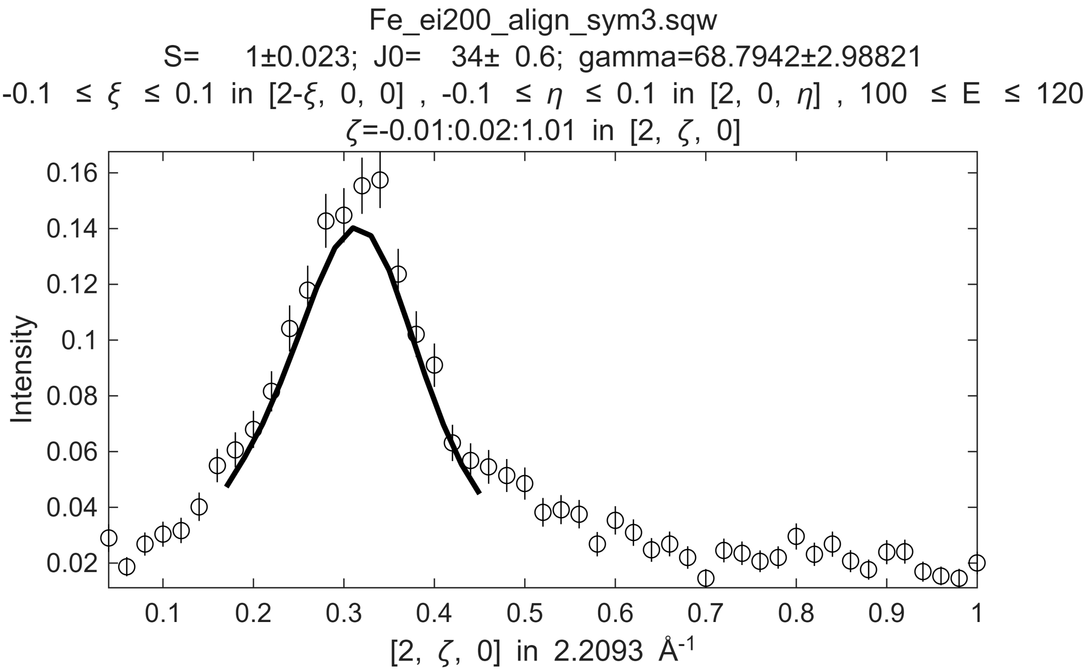

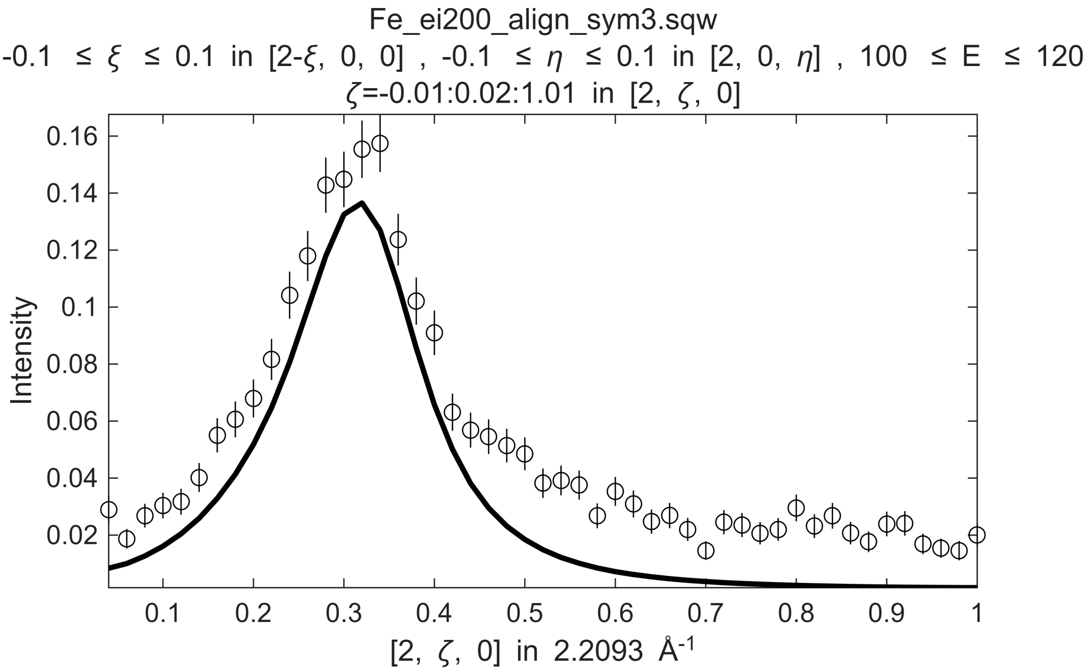


```
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*** step: 10#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:13 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.3173
```
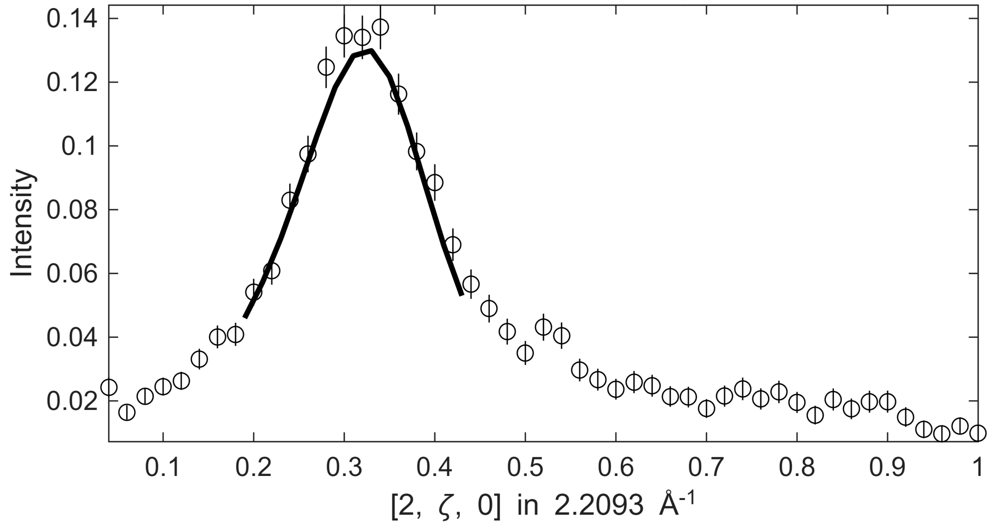
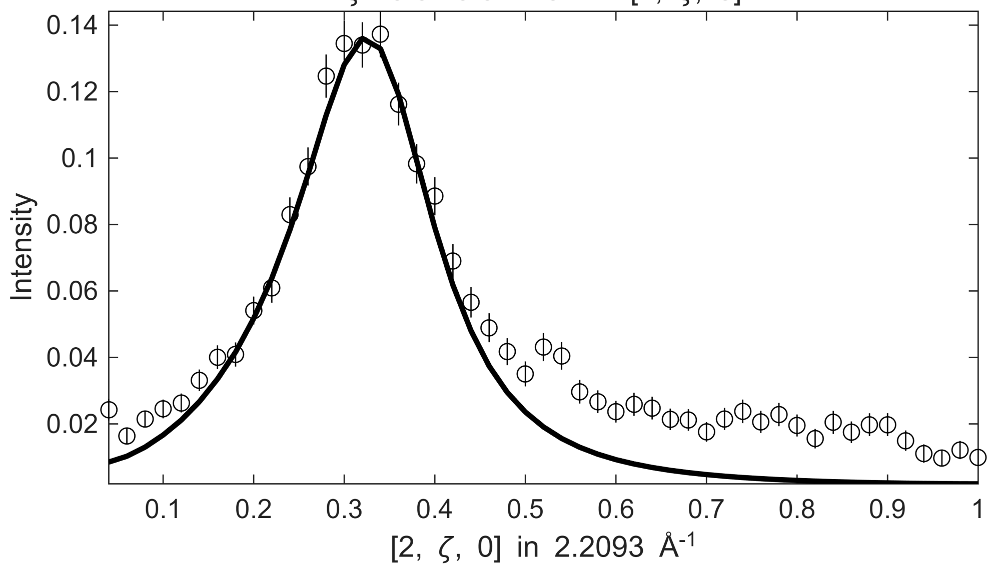

Fe_ei200_align_sym3.sqw
S= 1.2±0.021; J0= 36±0.51; gamma=83.3983±3.16611
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤ 130
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤ 130
ζ=-0.01:0.02:1.01 in [2, ζ, 0]




```
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*** step: 11#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.2478
```

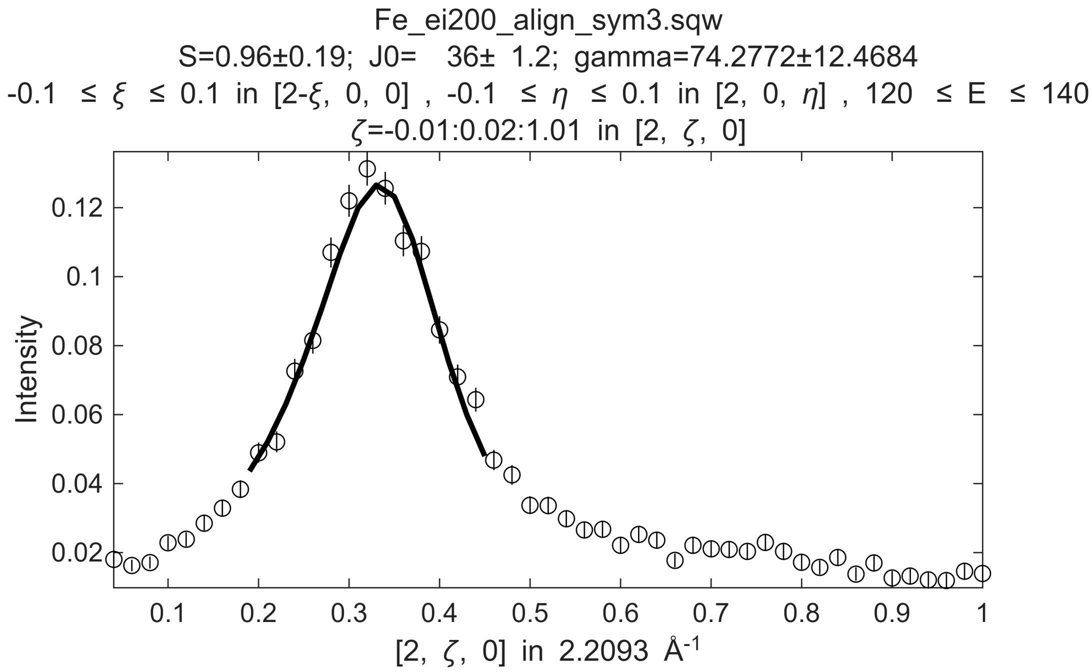

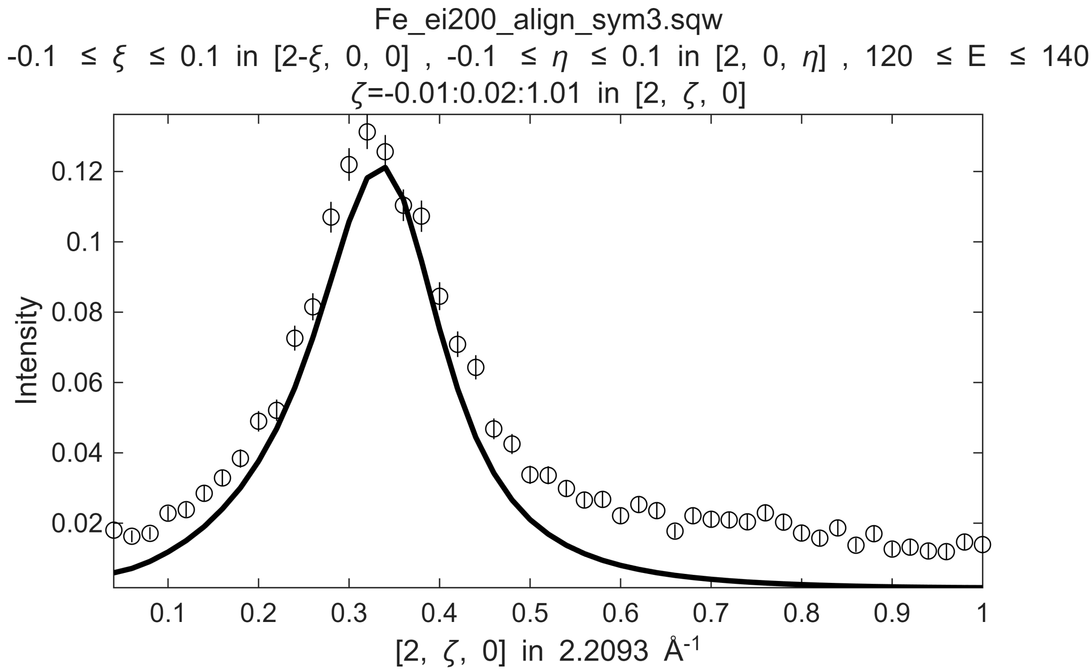


```
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*** step: 12#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:6 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.0695
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.64935
```

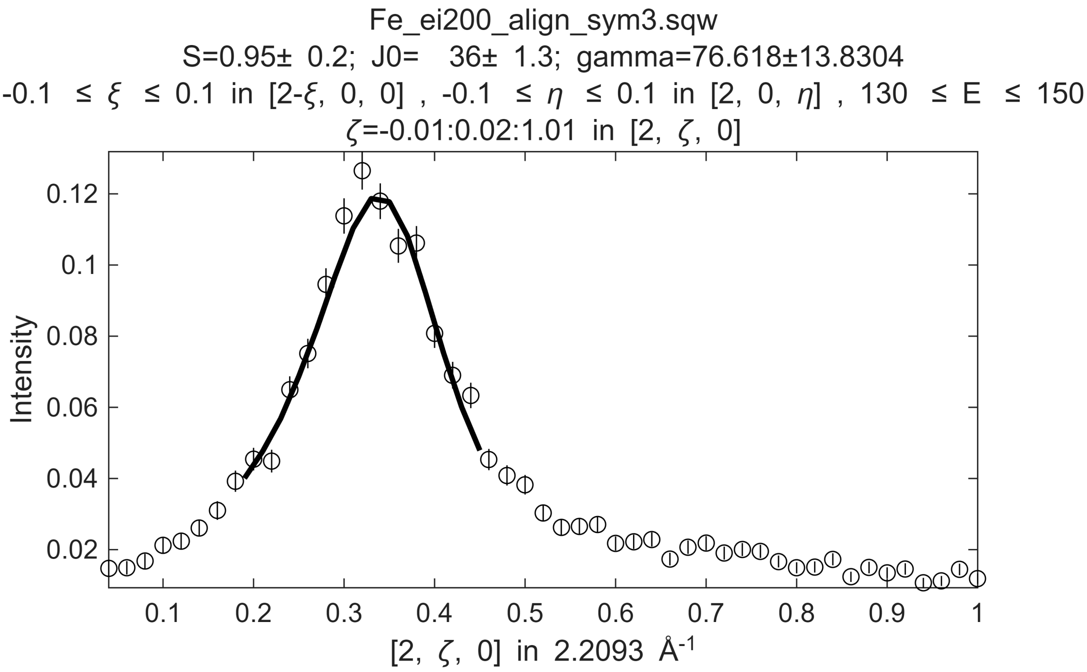

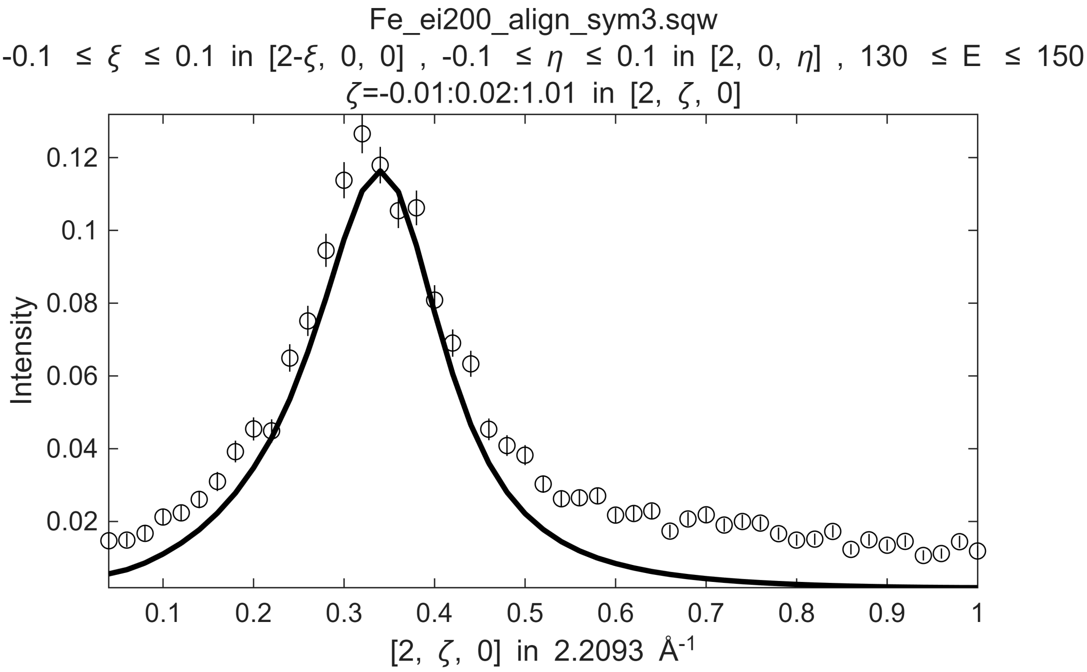


```
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*** step: 13#13  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
******************************************************************
Warning: Dataset:9 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.6043
```
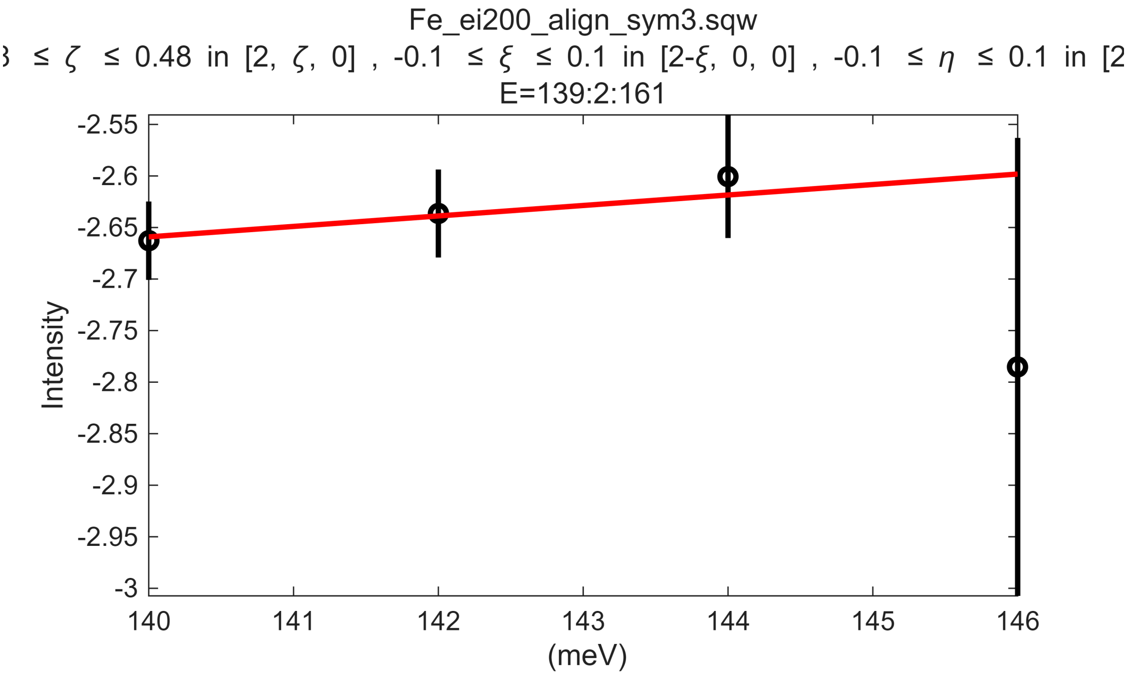
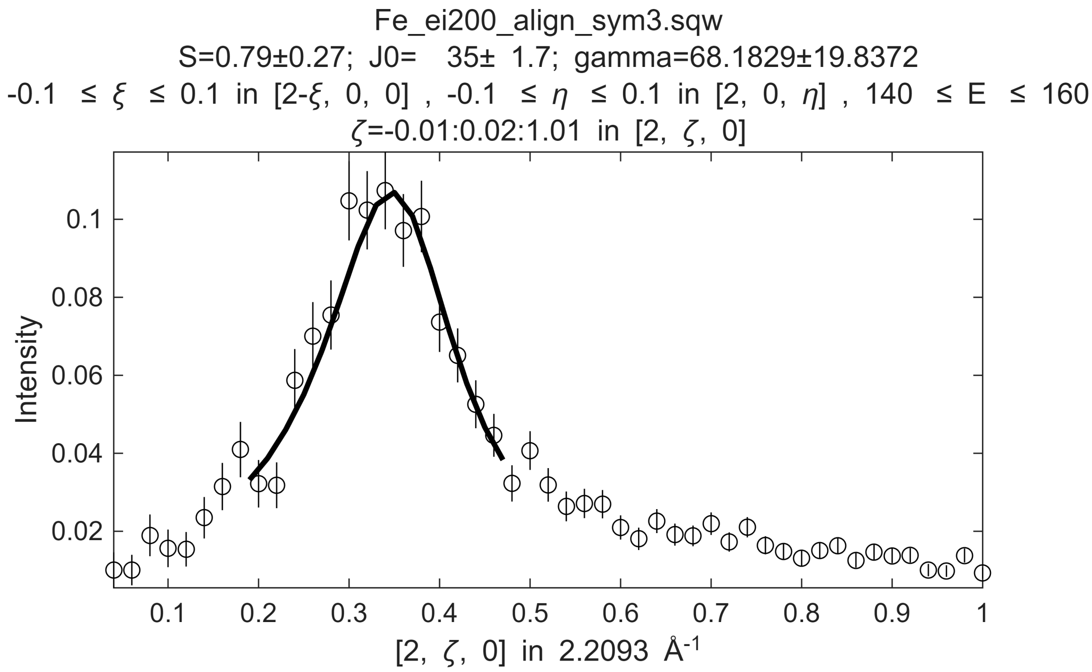
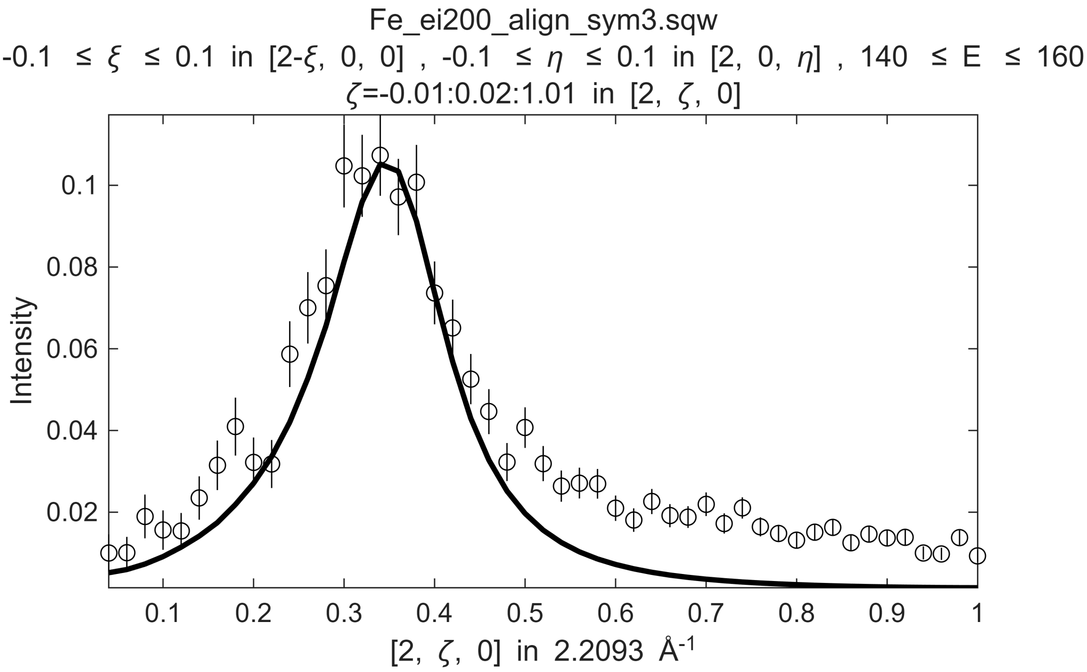




```
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.60606
```






```
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```
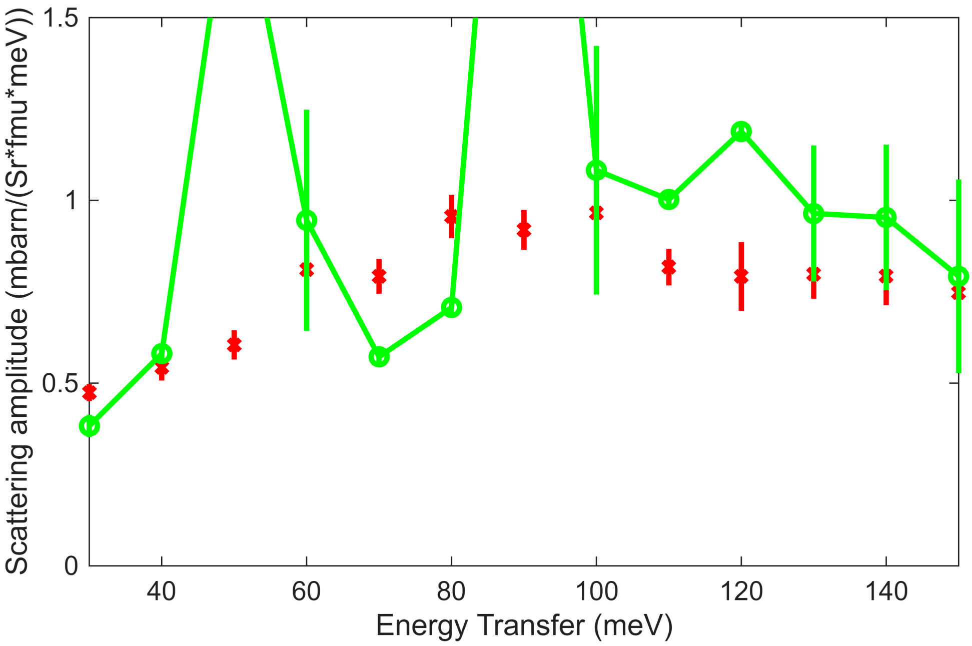
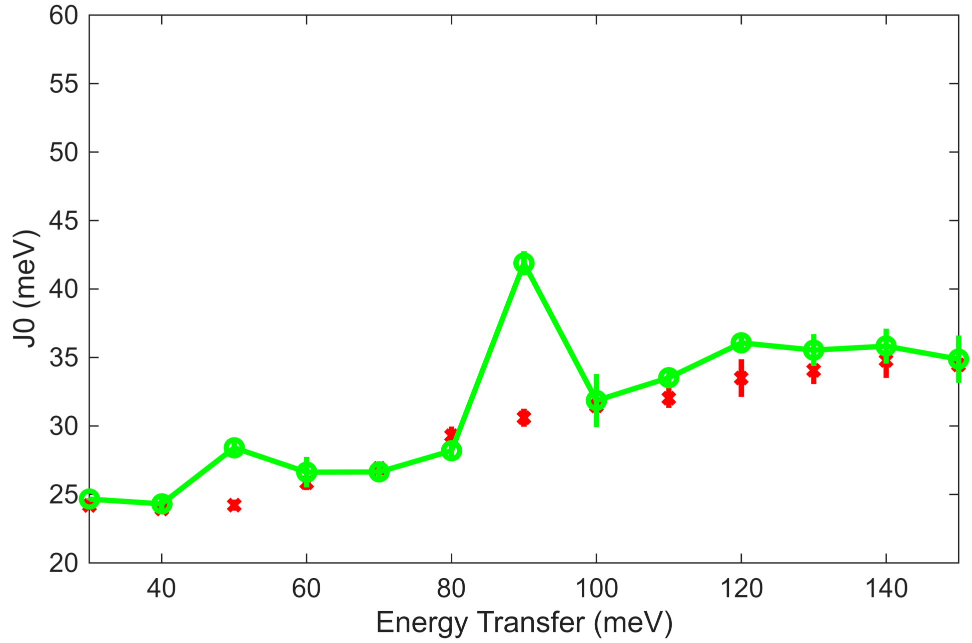
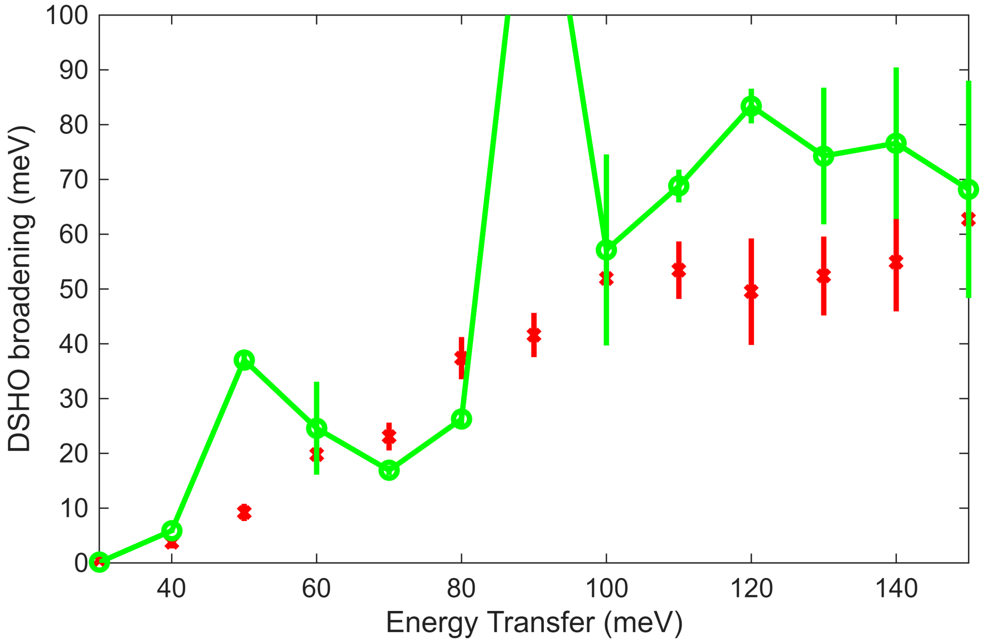
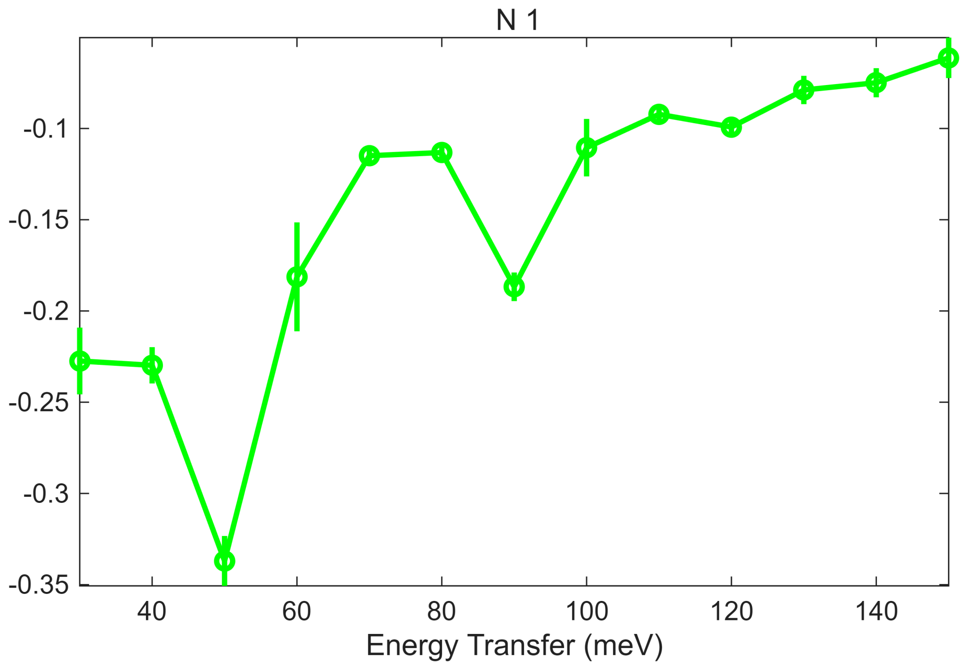

```
plot_SJG_results(fit_res,'g',{'S','J0','gamma'},{[0,1.5],[20,60],[0,100]},
{Sg,J0,G});
```



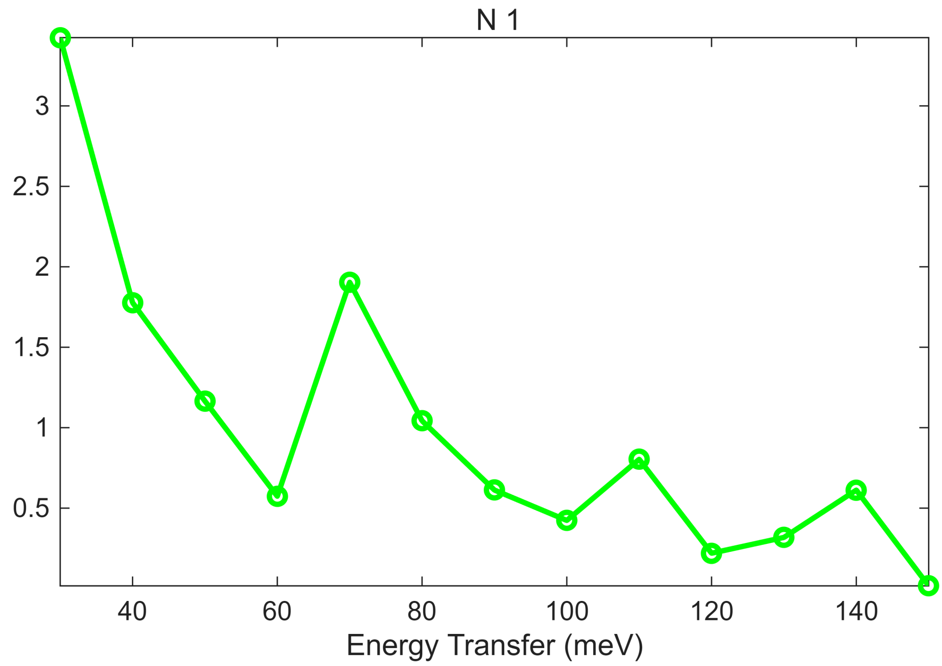




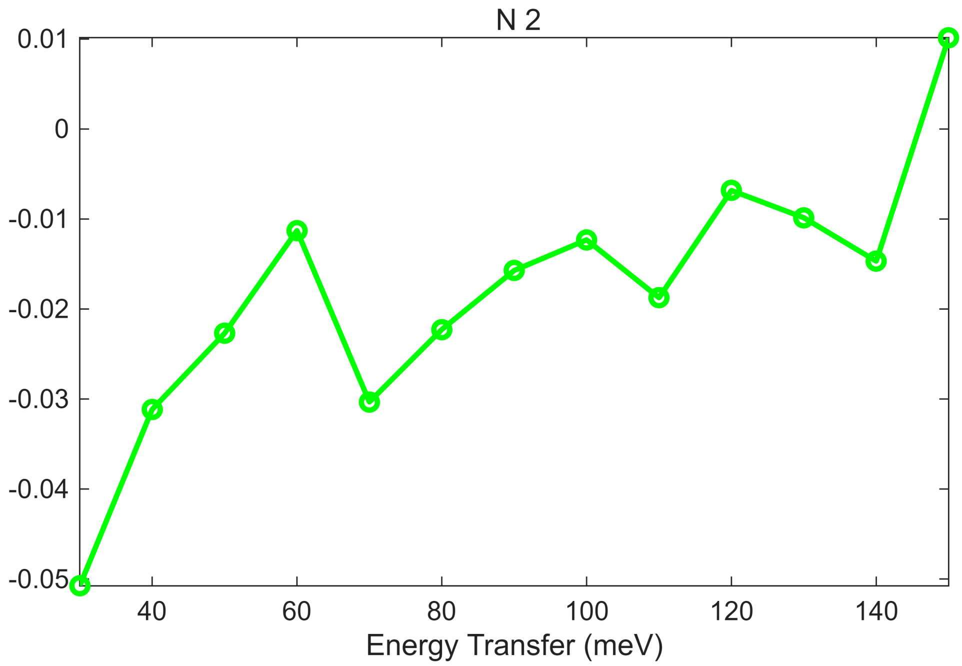

```
bg_data = extract_bg_par(fit_res.peak_fit_par,3);
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure;%ly -0.05 0.05
```




```
bg_data = extract_bg_par(fit_res.peak_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure;%ly -0.05 0.05
```




```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure;%ly -0.05 0.05
```




```
fit_res.Speak_loc_sl = Sg;
fit_res.J0peak_loc_sl = J0;
fit_res.gamma_peak_loc_sl = G;

% fit_res.Speak = Sg;
% fit_res.J0peak = J0;
% fit_res.gamma_peak = G;

fit_res.peak_fit_LocSlope_par = fit_res.peak_fit_par;
fit_resEi200 = fit_res;
save(fullfile(fit_data_path,fit_data_file),"fit_resEi200");
```